

# HAO QI

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## EDUCATION

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| <b>University of North Carolina at Charlotte (UNCC)</b>                                 | Charlotte, NC, USA  |
| • Ph.D. in Applied Mathematics (Statistics Concentration)                               | 08/2026 – Present   |
| <b>Rice University</b>                                                                  | Houston, TX, USA    |
| • Master of Statistics, GPA: 3.96/4.0                                                   | 08/2024 – 12/2025   |
| <b>University of North Carolina Wilmington (UNCW)</b>                                   | Wilmington, NC, USA |
| • Bachelor of Science in Mathematics (Statistics Minor) GPA: 3.94/4.0 (Dean's List)     | 08/2023 – 05/2024   |
| <b>Chongqing University of Arts and Sciences (CUAS)</b>                                 | Chongqing, China    |
| • Bachelor of Science in Mathematics and Applied Mathematics, GPA: 91.1/100 (Rank 1/95) | 09/2020 – 06/2024   |

## PUBLICATIONS

- Xinjie Lan, Yunbo Mei, Leheng Zhou, **Hao Qi**, Shichen Tang, Yunli Su, "Bayesian Time-aware Attention: Enhancing Transformers for Non-Stationary Time Series Forecasting", manuscript submitted to AAAI, 2026.
- Travis J. Miles, Michael T. Guinn, Xin Tan, **Hao Qi**, Vicente Orozco-Sevilla, Marc R. Moon, et al. "Tissue perfusion pressure: A novel hemodynamic measure to assess risk of acute kidney injury after cardiac surgery," *J. Thorac. Cardiovasc. Surg.*, 2026. DOI: 10.1016/j.jtcvs.2025.07.009.
- Hao Qi**, and Y.S. Wang, "Prediction and Analysis of Stock Logarithmic Returns Based on ARMA-GARCH Model", License: CC BY-NC 4.0, In book: Proceedings of CECNet 2022, DOI: 10.3233/FAIA220575
- Hao Qi**. "Research on Forecasting for Different Types of Small Sample Time Series Data", 2022/The 6th International Conferences on Data Science and Business Analytics (ICDSBA 2022). IEEE

## RESEARCH EXPERIENCE

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| <b>Longitudinal Variational Autoencoder for Within-Subject Image Prediction, Rice University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 02/2026 – Present |
| <i>Supervised by Dr. Huixia Judy Wang (Rice University)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |
| <ul style="list-style-type: none"><li>Developed a longitudinal variational autoencoder with a mixed-effects latent backbone and a first-order stochastic transition to model subject-specific temporal dynamics in image sequences.</li><li>Reformulated the task as prefix-conditioned missing-visit prediction, generating each visit from observed prior visits under a sequential latent variable framework.</li><li>Derived an ELBO-based objective for structured latent dynamics and evaluated one-step and recursive forecasting performance on longitudinal imaging data.</li></ul>                                                                                                                                                                                                                                                                                                                                   |                   |
| <b>A Novel Hemodynamic Measure to Assess Risk of Acute Kidney Injury After Cardiac Surgery, Rice &amp; BCM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 02/2025 – 07/2025 |
| <i>Supervised by Dr. Meng Li (Rice University)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |
| <ul style="list-style-type: none"><li>Applied univariate and multivariable logistic regression to quantify the association between tissue perfusion pressure (TPP) and postoperative acute kidney injury (AKI), adjusting for hemodynamic covariates including mean arterial pressure, with analyses stratified by vasoactive inotrope score (VIS).</li><li>Identified an optimal threshold of 38 mm Hg for TPP using regression-based risk estimation; below this level the odds of AKI increased markedly.</li><li>Demonstrated through odds ratio estimation and confidence interval inference that <math>TPP &lt; 38</math> mm Hg remained an independent predictor after statistical adjustment, including independence from mean arterial pressure in patients receiving vasoactive therapy.</li><li>Conducted prespecified sensitivity and subgroup analyses by VIS that yielded consistent effect estimates.</li></ul> |                   |

## WORK EXPERIENCE

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| <b>Rice University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Houston, TX, USA  |
| <i>Research Assistant (with Dr. Meng Li)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 01/2026 – 06/2026 |
| <ul style="list-style-type: none"><li>Developed dynamic machine learning models to predict acute kidney injury after cardiac surgery using hemodynamics, labs, medications, and patient clinical data.</li><li>Implemented rolling-window feature extraction and evaluated XGBoost, random forest, and logistic regression models under strict patient-level data splits.</li><li>Built calibration-based evaluation metrics, mapped predicted creatinine to KDIGO AKI stages, and conducted external validation using the MIMIC dataset.</li></ul> |                   |

## AWARDS AND HONORS

- China National Scholarship
- Chongqing Outstanding College Graduate Student

## SKILLS

- Programming Software: Python, R, MATLAB, SAS
- Office Tools: Word, Excel, PowerPoint, LaTeX
- Languages: Mandarin (native), English (senior)